Heewon Lee

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Research Experience

Programming Systems Laboratory, KAIST

Undergraduate Intern, Advisor: Prof. Kihong Heo | June 2023 – April 2024

- Co-developed **UnitCon**, a static-analysis-driven **targeted unit test synthesizer** for real-world Java programs.
- Prepared **open-source projects** for analysis, investigated generated tests to identify **previously unknown bugs**, and analyzed textbfdataflow paths to fix them.
- Co-authored the resulting paper, which received the **ACM SIGSOFT Distinguished Paper Award at FSE 2025**.

Independent Research

Advisor: Prof. Sukyoung Ryu | June 2025 – February 2026 (During Military Service)

- Formalized **asymptotic analysis** as a pedagogical **abstract interpretation**.
- Established soundness using a **Galois connection** and developed **interactive tools** for educational use.
- Selected for presentation at the **ACM SIGPLAN POPL 2026 Student Research Competition**.

Education

B.S. in Computer Science

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea
Double major in Physics | Minor in Materials Science and Engineering

- **GPA**: 4.10/4.30
- **Honors**: Dean's List (Top 3% in Department), Spring 2021, Spring 2023, Fall 2023
- **Anticipated Graduation**: August 2027
- Selected participant, Study Abroad Workshop, Korea Foundation for Advanced Studies (KFAS), Seoul, Korea | July 2026 – July 2027 (In Progress)

Republic of Korea Army

Signal Intelligence & Electronic Warfare Operator | May 2024 – November 2025

- Honorably discharged as Sergeant
- **Concurrent research** via remote guidance from Prof. Sukyoung Ryu

Publications & Presentations

UnitCon: Synthesizing Targeted Unit Tests for Java Runtime Exceptions.

S. Jang, Y. Ryou, ***H. Lee****, and K. Heo* [paper]
Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE), 2025, p.2053 – 2074.
ACM SIGSOFT Distinguished Paper Award

Asymptotic Analysis as an Abstract Interpretation.

H. Lee *and S. Ryu* [poster]
Poster presented at the ACM SIGPLAN POPL 2026 Student Research Competition.

Asymptotic Analysis: The Undergraduate's Abstract Interpretation.

H. Lee *and S. Ryu* [poster in Korean]
Poster presented at the Special Interest Group on Programming Languages (SIGPL) Winter School 2026.